IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

DISTRIBUTED OPTIMISATION AND LEARNING

Thursday, 9 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Vlaski
Second Marker(s) : K.K. Leung

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Final Exam (Spring 2024)

ELEC70082 – Distributed Optimisation and Learning

Instructor: Stefan Vlaski

09/05/2024, 10:00 – 13:00

Read all of the following information before starting the exam:

- The exam is open book. Feel free to use any written or printed material you have brought. You can use and refer to any results from the lecture notes that might be useful in answering the questions.
- Calculators or other electronic devices are not allowed for this exam.
- There is a total of 3 problems that total to 100 marks. There is a total of 4 pages.
- You have 3 hours (10:00 - 13:00) to complete the exam.

Question 1. The following are true/false questions. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) [8 marks] (true/false) The consensus + innovations algorithm is bias-free when agents are homogenous, i.e., when $w_k^o = w^o$ for all k.

(b) [8 marks] (true/false) Suppose you are provided with a doubly-stochastic combination matrix $A \in \mathbb{R}^{10\times 10}$, and find its set of eigenvalues:

$$\{1, 1, 0.9, 0.9, 0.5, 0.5, 0.4, 0.3, 0.2, 0.1\} \tag{1}$$

From this, you can conclude that the graph is strongly-connected.

(c) [8 marks] (true/false) In the presence of gradient noise, the difference in performance between penalty-based and primal-dual algorithms becomes negligible for very small step-sizes.

(d) [8 marks] (true/false) The implementation of a gradient-tracking based algorithm for decentralized optimisation requires a larger number of communication exchanges than a primal-dual or penalty-based algortihm.

(e) [8 marks] (true/false) Consider 2 different learning setups. In setup one, a single agent is running a stochastic gradient algorithm using mini-batch gradient approximations with $B = 100$. In setup two, $K = 50$ agents are running a consensus+innovations algorithm using mini-batch gradient approximations with $B = 2$. Otherwise, all agents, problem and learning parameters are identical, data is independently and identically distributed, and the step-size is small enough so that higher-order terms can be neglected. Then, the performance of the two setups in terms of their steady-state error will be approximately the same.

Question 2. We consider a collection of K agents, where each agent k is equipped with a local logistic risk:

$$J_k(w) \triangleq \mathbb{E} \ln \left(1 + e^{-\boldsymbol{\gamma}_{k,i} \boldsymbol{h}_{k,i}^{\mathsf{T}} w}\right) + \frac{\rho}{2}\|w\|^2 \tag{2}$$

We are interested in solving the consensus optimization problem:

$$w^o \triangleq \arg \min_{w \in \mathbb{R}^M} \frac{1}{K} \sum_{k=1}^{K} J_k(w) \tag{3}$$

(a) [10 marks] Suppose each agent sees B observations $\{\boldsymbol{h}_{k,i,b}, \boldsymbol{\gamma}_{k,i,b}\}_{b=1}^{B}$ at time i. All observations (across space and time) are independent. Formulate the recursions for the diffusion and for the exact diffusion algorithm, when agents are employing local mini-batch approximations of the stochastic gradient, each with size B.

(b) [10 marks] Determine a bound for the limiting performance $\limsup_{i\to\infty} \mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters, strong-convexity and Lipschitz constants of $J(w)$, locally optimal models w_k^o, mixing rate λ_2 and data statistics for both the diffusion and the exact diffusion algorithm.

(c) [5 marks] Simplify the expressions to the case where agents are homogenous and $w_k^o = w^o$. Compare the performance and comment.

Question 3. We examine a variant of the consensus averaging algorithm over noisy links:

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} a_{\ell k} \boldsymbol{w}_{\ell,i-1} + \boldsymbol{v}_{k,i} \tag{4}$$

Each agent is initialized at some deterministic local signal g_k, i.e., $w_{k,0} = g_k$. The combination weights are collected in a matix A that satisfies:

$$A = A^{\mathsf{T}} \tag{5}$$

$$A\mathbb{1} = \mathbb{1} \tag{6}$$

The communication noise $\boldsymbol{v}_{k,i}$ is modelled as zero-mean with $\mathbb{E}\|\boldsymbol{v}_{k,i}\|^2 = \sigma_{v_k}^2$. You may assume $\boldsymbol{v}_{k,i}$ to be independent across agents, and the iterates $\boldsymbol{w}_{k,i-1}$.

(a) [10 marks] Determine a recursion for the evolution of the network centroid:

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K} \sum_{k=1}^{K} \boldsymbol{w}_{k,i} \tag{7}$$

(b) [15 marks] Determine a bound for the expected squared deviation of the local estimates $\boldsymbol{w}_{k,i}$ from the centroid $\boldsymbol{w}_{c,i}$ as $i \to \infty$, captured in the quantity:

$$\limsup_{i \to \infty} \sum_{k=1}^{K} \mathbb{E}\|\boldsymbol{w}_{k,i} - \boldsymbol{w}_{c,i}\|^2 \tag{8}$$

(c) [10 marks] Determine an expression for the expected squared deviation of the centroid from average, i.e., for:

$$\mathbb{E} \left\| \boldsymbol{w}_{c,i} - \frac{1}{K} \sum_{k=1}^{K} g_k \right\|^2 \tag{9}$$

Based on your conclusion, what can you say about the robustness of the average consensus algorithm to noisy links?